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Second South Asia Edition

Sturdevant's ART AND SCIENCE OF OPERATIVE DENTISTRY



50 Years of Publishing
1968–2018

André V. Ritter • Lee W. Boushell • Ricardo Walter

Adaptation Editor
V. Gopikrishna

Restricted Edition

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Sturdevant's Art and Science of Operative Dentistry

Sturdevant's Art and Science of Operative Dentistry

Second South Asia Edition

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This book is dedicated to

My Parents

Sulochana… for being my beloved Amma (mother) and my Alpha (beginning of life)…
Ambuja… for being the Annapoorni (nurturer) of our lives…
M Velayutham… for teaching me integrity and humility in life…
VG Sivasubramanian… for showing me the path of caring and sharing…

My Teachers

Dr E Munirathnam Naidu… for showing me the attributes of discipline, hard work and perseverance…
Professors A Parameswaran and *B Suresh Chandra…* for instilling in me the drive to learn, teach and research…
Dr K Sridhar… for making me aware that water always finds its level…
Dr Vijailakshmi Acharya… for inspiring me to be passionate about perfection…

My Family

And above all to… ***Grace…*** for being the wind beneath my wings…

—V. Gopikrishna

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Foreword

Dr Clifford Sturdevant had a brass plaque on his desk that reads *'If it's almost right it's wrong!'* This commitment to excellence also was the mantra upon which his classic textbook, *The Art and Science of Operative Dentistry*, was first written and published in 1968. This textbook has been the basis for training dental students in the fine art and clinical science of operative dentistry for 50 years. In light of this significant landmark, which coincides with the publication of this new seventh edition, we believe it is important to present the evolution of the various editions of the textbook from a historical perspective.

The first edition (Sturdevant, Barton, Brauer, 1968) was meant 'to present the significant aspects of operative dentistry and the research findings in the basic and clinical sciences that have immediate application' in the field of operative dentistry. It is important to note that Dean Brauer pointed out in his preface that beyond having the knowledge and skills needed to perform a procedure, the practitioner must also have high moral and ethical standards, essential and priceless ingredients. Since the first edition, this textbook series has always attempted to present artistic and scientific elements of operative dentistry in the context of ethical standards for patient care.

It is also worth noting that the first edition was printed and bound in 'landscape' format so that it could more easily be used as a manual in the preclinical laboratory and would always remain open to the desired page. The handmade 5X models used to illustrate the various steps in cavity preparation were created by two dental students enrolled at Chapel Hill School of Dentistry, the University of North Carolina, during the writing of the first edition. Illustrations of these models have continued to be used in later editions, and the models themselves have served as important teaching materials for decades.

Although the techniques, materials, armamentarium, and treatment options continue to evolve, many of the principles of operative dentistry described in the first edition are still pertinent today. An understanding of these principles and the ability to meticulously apply them are critical to provide the outstanding dental treatment expected by our patients.

The *second edition (Sturdevant, Barton, Sockwell, Strickland, 1985)* expanded on many techniques (e.g. acid etching) using experience and published research that had occurred since publication of the first edition. The basics of occlusion were emphasized and presented in a way that would be helpful to the dental student and practitioner. A chapter on treatment planning and sequencing of procedures, as well as a chapter providing a thorough treatise on the use of pins, was included. Information on silicate cement, self-curing acrylic resin and the baked porcelain inlay was eliminated for obvious reasons. The chapters on endodontic therapy and the 'dental assistant' were no longer included. Chapters on (1) tooth-coloured restorations and (2) additional conservative and esthetic treatments explained the changes and improvements that occurred in the areas of esthetic options available to patients. In the chapter on gold inlay/onlay restorations, increased emphasis was given to the gold onlay restorations for Class II cavity preparations.

The *third edition (Sturdevant, Roberson, Heymann, J. Sturdevant, 1995)* placed a new emphasis on cariology and the 'medical model of disease' with regard to risk assessment and managing the high-risk caries patient. This important concept laid

the foundation for what is still taught today with regard to identifying risk factors and defining a treatment plan based on caries risk assessment. The third edition also included new expanded chapters on infection control, diagnosis and treatment planning and dental materials. In light of the growing interest in the area of esthetic dentistry, a variety of conservative esthetic treatments were introduced including vital bleaching, micro- and macro-abrasion, etched porcelain veneers and the novel all-porcelain bonded pontic. Additionally, an entirely new section on tooth-coloured inlays and onlays was included that chronicled both lab-processed resin and ceramic restorations of this type and those fabricated chairside with CAD/CAM systems.

With the *fourth edition* of this text *(Roberson, Heymann, Swift, 2002)*, Dr Clifford Sturdevant's name was added to the book title to honour his contributions to the textbook series and the discipline of operative dentistry. In this edition, a particular emphasis was placed on bonded esthetic restorations. Consequently, an entirely new chapter was included on fundamental concepts of enamel and dentin adhesion. This chapter was intended to provide foundational information critical to the long-term success of all types of bonded restorations.

The *fifth edition (Roberson, Heymann, Swift, 2006)* continued with the renewed emphasis on the importance of adhesively bonded restorations and focused on scientific considerations for attaining optimal success, particularly with posterior composites. Concepts such as the 'C Factor' and keys to reducing polymerization effects were emphasized along with factors involved in reducing micro-leakage and recurrent decay.

The sixth edition *(Heymann, Swift, Ritter, 2013)* represented a transition from a large printed edition, as in the past, to a smaller, streamlined printed version that focused on concepts and techniques immediately essential for learning contemporary operative dentistry. The same amount of information was included, but many chapters such as those addressing biomaterials, infection control, pain control, bonded splints and bridges, direct gold restorations and instruments and equipment were available for the first time in a supplemental online format.

The *seventh edition (Ritter, Boushell, Walter 2018)* continues the legacy of maintaining and enriching the fundamental concepts and principles of contemporary operative dentistry. Diagnosis, classification and management of dental caries have been significantly updated in light of the latest clinical and epidemiological research. Similarly, content on adhesive dentistry and composite resins has been updated as a result of the evolving science in these fields.

An entirely new chapter on light curing and its important role in the clinical success of resin composite restorations has been added. Moreover, a new scientifically based chapter details the important elements of colour and shade matching and systematically reviews how the dental clinician is better able to understand the many covariables involved in colour assessment. It also reviews how best to improve shade matching of esthetic restorations to tooth structure.

In an attempt to better optimize restorative treatment outcomes involving periodontal challenges, a new chapter has been included that addresses these principles. Periodontology applied to operative dentistry chronicles the various clinical considerations involving conditions such as inadequate crown length, lack of root coverage and other vexing problems requiring interdisciplinary treatment to optimize success.

Finally, this text addresses the ever-evolving area of digital dentistry with a new chapter, Digital Dentistry in Operative Dentistry. This chapter reviews the various technologies involved in scanning and image capture for both treatment planning and restorative applications. Additionally, the authors review various types of digital restorative systems for both chairside and modem-linked laboratory-based fabrication of restorations. In recognition of the rapid movement to digital dentistry, this chapter is a vital addition to a textbook whose tradition has been always to reflect the latest technologies and research findings in contemporary operative dentistry.

Since its inception 50 years ago, the Sturdevant text has been a dynamic document, with content that has included innovative information on the latest materials and techniques. Over this time period, numerous internationally recognized experts

have addressed many specific topics as authors and coauthors of various chapters. It also should be pointed out that with all editions of the textbook, the authors of the various chapters are themselves actively involved in teaching students preclinical and clinical operative dentistry. Moreover, they are 'wet-fingered dentists' who also practice operative dentistry for their individual patients.

In summary, for 50 years Sturdevant's Art and Science of Operative Dentistry has been a major resource guiding educators in the teaching of contemporary operative dentistry. Each edition of this text has striven to incorporate the latest technologies and science based on the available literature and supporting research. This edition is a superb addition to this tradition, which will most assuredly uphold the standard of publication excellence that has been the hallmark of the Sturdevant textbooks for half a century.

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Preface

to the Second South Asia Edition



Dr Clifford Sturdevant

"If you would not be forgotten as soon as you are dead…

Either write something worth reading or do something worth writing…"

—Benjamin Franklin

The first edition of *Sturdevant's Art and Science of Operative Dentistry* was published in 1968 and in these past fifty years it has always been considered to be the *Bible of Operative Dentistry* as it is the most comprehensive operative dentistry textbook. Drawing from both theory and practice and supported by extensive clinical and laboratory research, it presents a clearly detailed, heavily illustrated step-by-step approach to conservative, restorative and preventive dentistry.

The *first South Asian Adaptation Edition* of this iconic textbook was brought out in 2013 and this current second edition of the same is an endeavour to carry on the legacy.

Based upon the principle that dental caries is a disease, the book provides both a thorough understanding of caries and an authoritative approach to its treatment and prevention. Throughout the book, emphasis is placed on the importance of treating the underlying causes of the patient problem(s), not just restoring the damage that has occurred. It is organized in a sequential format; the early chapters present the necessary general information while the later chapters are specifically related to the practice of operative dentistry, including conservative esthetic procedures.

This Second South Asia edition of *Sturdevant's Art and Science of Operative Dentistry* has been **significantly revised** in order to streamline the text and improve readability. The order and content of chapters have been reorganized keeping in mind the needs of both undergraduate and postgraduate students.

Five new chapters, namely '***Periodontology Applied to Operative Dentistry***', '***Colour and Shade Matching in Operative Dentistry***', '***Light Curing in Operative Dentistry***', '***Digital Dentistry in Operative Dentistry***' and '***Resin-Bonded Splints and Bridges***' have been included in this edition. In addition, the book is now in **full colour**. The line art for the book has been completely redrawn in full colour to better show techniques and details, and new, full colour photos have been added where appropriate, and highlighted important concepts and clinical tips for the benefit of the student and clinician. To publish this edition on the year we commemorate the 50th anniversary of the publication of the First Edition is a milestone for Operative Dentistry. I am honored to have had the opportunity to work on and present the Second South Asia Edition.

V. Gopikrishna

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V. Gopikrishna

Page left intentionally blank

Brief Contents

Foreword ..... *vii*
Contributors ..... *xi*
Preface to the Second South Asia Edition ..... *xiii*
Acknowledgements ..... *xv*

1. Clinical Significance of Dental Anatomy, Histology, Physiology and Occlusion 1
2. Dental Caries: Etiology, Clinical Characteristics, Risk Assessment and Management 25
3. Patient Assessment, Examination, Diagnosis, and Treatment Planning 83
4. Infection Control 111
5. Pain Control for Operative Dentistry 125
6. Dental Biomaterials 131
7. Instruments and Equipment for Tooth Preparation 179
8. Preliminary Considerations for Operative Dentistry 207
9. Dentin Hypersensitivity 239
10. Periodontology Applied to Operative Dentistry 245
11. Fundamentals of Tooth Preparation and Pulp Protection 263
12. Introduction to Silver Amalgam 285
13. Clinical Technique for Amalgam Restorations 301
14. Complex Amalgam Restorations 375
15. Fundamental Concepts of Enamel and Dentin Adhesion 401
16. Introduction to Composite Resins 435
17 Colour and Shade Matching in Restorative Dentistry 445
18. Light Curing of Restorative Materials 465
19. Clinical Technique for Direct Composite Resin and Glass Ionomer Restorations 495
20. Non-carious Lesions and Their Management 547
21. Additional Conservative Esthetic Procedures 559
22. Digital Dentistry in Operative Dentistry 611
23. Class II Cast Metal Restorations 633
24. Direct Gold Restorations 671
25. Resin-bonded Splints and Bridges 679

Index *695*

Supplemental Online Chapters

26. Class III and V Amalgam Restorations e1
27. Additional Information on Instruments and Equipment for Tooth Preparation e21

Page left intentionally blank